

Three Independent Limit Curves

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1 TCP Velocity Limit Curve Derivation

1.1 Kinematic Relationships

Path parameterization:

$$\mathbf{q} = \mathbf{f}(s), \quad 0 \leq s \leq s_f \quad (1)$$

Joint velocities:

$$\dot{\mathbf{q}} = \mathbf{f}'(s)\dot{s} \quad (2)$$

TCP velocity via forward kinematics:

$$\mathbf{v}_{\text{TCP}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}} \quad (3)$$

Substituting:

$$\mathbf{v}_{\text{TCP}} = \mathbf{J}(\mathbf{f}(s))\mathbf{f}'(s)\dot{s} \quad (4)$$

1.2 TCP Speed Constraint

The actual TCP velocity is a 3D vector:

$$\mathbf{v}_{\text{TCP}} = \mathbf{J}(\mathbf{f}(s))\mathbf{f}'(s)\dot{s} \quad (5)$$

The TCP velocity **direction** is determined by $\mathbf{J}(\mathbf{f}(s))\mathbf{f}'(s)$ (path geometry and robot configuration). We control only the **magnitude** (speed) by adjusting $\dot{s}$.

Taking magnitudes (with $\dot{s} > 0$):

$$\|\mathbf{v}_{\text{TCP}}\| = \|\mathbf{J}(\mathbf{f}(s))\mathbf{f}'(s)\|\dot{s} \quad (6)$$

To limit TCP speed to v_{TCP} (the desired input value):

$$\|\mathbf{v}_{\text{TCP}}\| \leq v_{\text{TCP}} \quad (7)$$

Substituting:

$$\|\mathbf{J}(\mathbf{f}(s))\mathbf{f}'(s)\|\dot{s} \leq v_{\text{TCP}} \quad (8)$$

1.3 TCP Limit Curve

Solving for $\dot{s}$:

$$\boxed{\dot{s}_{\text{max}}^{\text{TCP}}(s) = \frac{v_{\text{TCP}}}{\|\mathbf{J}(\mathbf{f}(s))\mathbf{f}'(s)\|}} \quad (9)$$

2 Computing the TCP Limit Curve

2.1 Required Components

2.1.1 Path Configuration

$$\mathbf{q} = \mathbf{f}(s) \quad (10)$$

2.1.2 Path Derivative

Linear segments:

$$\mathbf{f}'(s) = \frac{\mathbf{q}_{i+1} - \mathbf{q}_i}{\|\mathbf{q}_{i+1} - \mathbf{q}_i\|} \quad (11)$$

Circular blends (Kunz-Stilman Eq. 8):

$$\mathbf{f}'(s) = -\hat{\mathbf{x}}_i \sin\left(\frac{s}{r_i}\right) + \hat{\mathbf{y}}_i \cos\left(\frac{s}{r_i}\right) \quad (12)$$

2.1.3 Jacobian Matrix

Linear velocity Jacobian:

$$\mathbf{J}(\mathbf{q}) = [\mathbf{J}_1(\mathbf{q}) \quad \mathbf{J}_2(\mathbf{q}) \quad \cdots \quad \mathbf{J}_n(\mathbf{q})] \in \mathbb{R}^{3 \times n} \quad (13)$$

For revolute joint i :

$$\mathbf{J}_i(\mathbf{q}) = \mathbf{z}_{i-1} \times (\mathbf{p}_{\text{TCP}} - \mathbf{p}_{i-1}) \quad (14)$$

2.1.4 Kinematic Gain

$$\mathbf{g}(s) = \mathbf{J}(\mathbf{f}(s))\mathbf{f}'(s) \in \mathbb{R}^3 \quad (15)$$

$$\text{gain}(s) = \|\mathbf{g}(s)\| = \sqrt{g_x^2 + g_y^2 + g_z^2} \quad (16)$$

2.1.5 Limit Curve Value

$$\dot{s}_{\text{max}}^{\text{TCP}}(s) = \frac{v_{\text{TCP}}}{\text{gain}(s)} \quad (17)$$

2.2 Singularity Handling

When $\|\mathbf{J}\mathbf{f}'\| \rightarrow 0$, the TCP limit goes to infinity:

$$\dot{s}_{\text{max}}^{\text{TCP}}(s) = \begin{cases} \frac{v_{\text{TCP}}}{\|\mathbf{J}(\mathbf{f}(s))\mathbf{f}'(s)\|} & \text{if } \|\mathbf{J}(\mathbf{f}(s))\mathbf{f}'(s)\| > \epsilon \\ \infty & \text{otherwise} \end{cases} \quad (18)$$

where $\epsilon = 10^{-6}$.

3 Three Independent Limit Curves

3.1 Overall Limit Curve

The Kunz-Stilman algorithm has two limit curves:

$$\dot{s}_{\max}^{\text{acc}}(s) \quad (\text{acceleration limit, Eq. 31}) \quad (19)$$

$$\dot{s}_{\max}^{\text{vel,joint}}(s) \quad (\text{joint velocity limit, Eq. 36}) \quad (20)$$

We add a third independent limit curve:

$$\dot{s}_{\max}^{\text{TCP}}(s) \quad (\text{TCP velocity limit}) \quad (21)$$

The overall limit is:

$$\dot{s}_{\max}(s) = \min \left(\dot{s}_{\max}^{\text{acc}}(s), \dot{s}_{\max}^{\text{vel,joint}}(s), \dot{s}_{\max}^{\text{TCP}}(s) \right) \quad (22)$$

3.2 Active Constraint Identification

At any position s , the active constraint is:

$$\text{Active}(s) = \arg \min \left(\dot{s}_{\max}^{\text{acc}}(s), \dot{s}_{\max}^{\text{vel,joint}}(s), \dot{s}_{\max}^{\text{TCP}}(s) \right) \quad (23)$$

4 Algorithm Modifications

4.1 Modified Step 2

The forward integration step checks against all three limit curves.

Step 2: Integrate forward with $\ddot{s} = \ddot{s}_{\max}(s, \dot{s})$ until one of the following conditions is met:

- If $s \geq s_f$: go to step 5
- If $\dot{s} > \dot{s}_{\max}^{\text{acc}}(s)$: go to step 4
- If $\dot{s} > \dot{s}_{\max}^{\text{vel,joint}}(s)$: go to step 3a
- If $\dot{s} > \dot{s}_{\max}^{\text{TCP}}(s)$: go to step 3b *(new)*

4.2 Step 3a: Follow Joint Velocity Limit

Step 3a: Follow joint velocity limit curve $\dot{s}_{\max}^{\text{vel,joint}}(s)$ until one of the following conditions is met:

- If $\frac{\ddot{s}_{\max}(s, \dot{s}_{\max}^{\text{vel,joint}}(s))}{\dot{s}} < \frac{d}{ds} \dot{s}_{\max}^{\text{vel,joint}}(s)$: go to step 2
- If $\frac{\ddot{s}_{\min}(s, \dot{s}_{\max}^{\text{vel,joint}}(s))}{\dot{s}} > \frac{d}{ds} \dot{s}_{\max}^{\text{vel,joint}}(s)$: go to step 4

4.3 Step 3b: Follow TCP Velocity Limit (New)

Step 3b: Follow TCP velocity limit curve $\dot{s}_{\max}^{\text{TCP}}(s)$ until one of the following conditions is met:

- If $\frac{\ddot{s}_{\max}(s, \dot{s}_{\max}^{\text{TCP}}(s))}{\dot{s}} < \frac{d}{ds} \dot{s}_{\max}^{\text{TCP}}(s)$: go to step 2
- If $\frac{\ddot{s}_{\min}(s, \dot{s}_{\max}^{\text{TCP}}(s))}{\dot{s}} > \frac{d}{ds} \dot{s}_{\max}^{\text{TCP}}(s)$: go to step 4

4.4 Remaining Steps Unchanged

Steps 1, 4, and 5 remain unchanged from Kunz-Stilman Section VI.

5 Switching Point Analysis for TCP Limit Curve

Switching points on the TCP velocity limit curve are found using the same methods as for joint velocity limits (Kunz-Stilman Section VII-B).

5.1 Continuous Case

Following Kunz-Stilman Section VII-B-1, a point s on the TCP limit curve is a potential switching point if:

$$\ddot{s}_{\min}(s, \dot{s}_{\max}^{\text{TCP}}(s)) = \frac{d}{ds} \dot{s}_{\max}^{\text{TCP}}(s) \quad (24)$$

These switching points are found numerically using bisection.

5.2 Derivative of TCP Limit Curve

The derivative is:

$$\frac{d}{ds} \dot{s}_{\max}^{\text{TCP}}(s) = -v_{\text{TCP}} \frac{\frac{d}{ds} \|\mathbf{J}\mathbf{f}'\|}{\|\mathbf{J}\mathbf{f}'\|^2} \quad (25)$$

where:

$$\frac{d}{ds} \|\mathbf{J}\mathbf{f}'\| = \frac{(\mathbf{J}\mathbf{f}')^T \cdot \frac{d}{ds} [\mathbf{J}\mathbf{f}']}{\|\mathbf{J}\mathbf{f}'\|} \quad (26)$$

and:

$$\frac{d}{ds} [\mathbf{J}\mathbf{f}'] = \frac{\partial \mathbf{J}}{\partial \mathbf{q}} \mathbf{f}' \mathbf{f}' + \mathbf{J} \mathbf{f}'' \quad (27)$$

5.3 Discontinuous Case

TCP limit discontinuities occur when $\mathbf{f}''(s)$ is discontinuous (e.g., at transitions between linear segments and circular blends).

Following Kunz-Stilman Section VII-B-2, switching points at discontinuities are identified by evaluating $\ddot{s}_{\min}$ on both sides of the discontinuity.

A discontinuity at s is a switching point if the conditions in Kunz-Stilman Eq. 41-42 are satisfied when evaluated with $\dot{s}_{\max}^{\text{TCP}}(s)$.

5.4 Numerical Search Implementation

The bisection search for TCP velocity switching points follows the same procedure as for joint velocity limits:

1. Step along the TCP limit curve $\dot{s}_{\max}^{\text{TCP}}(s)$

2. At each step, evaluate:

$$\Delta(s) = \ddot{s}_{\min}(s, \dot{s}_{\max}^{\text{TCP}}(s)) - \frac{d}{ds} \dot{s}_{\max}^{\text{TCP}}(s) \quad (28)$$

3. When a sign change in $\Delta(s)$ is detected, use bisection to locate the switching point
4. Verify the switching point satisfies the conditions in Step 3b